Royal Crest Academy
SCIENCE · INNOVATION · SKILL

FREE LEARNER GUIDE

Technical & Vocational *study guide*

Technical and vocational subjects turn knowledge into practical, employable skill, aligned to Zimbabwe's technical education framework.

Big ideas

Safety first

Every practical task starts with the right protective equipment, safe tools and a clean, organised work space.

Plan, make, check

Good work follows a cycle: read the drawing, plan the steps, make it accurately, then check it against the specification.

Skills that earn

Practical skills lead to apprenticeships, small businesses and further study in engineering, design and hospitality.

Topic checklist

tick each topic when you can explain it

SKILL AREAS

- [ ] Technical graphics and drawing
- [ ] Reading orthographic and isometric drawings
- [ ] Measuring and marking out
- [ ] Hand tools and their safe use
- [ ] Joining materials
- [ ] Finishing and presentation
- [ ] Costing a product
- [ ] Quality control

SKILLS CLUBS

- [ ] Welding & Aluminium Fabrication
- [ ] Cake Making
- [ ] Event Décor & Interior Décor
- [ ] Beauty Therapy (Pedicure & Manicure)

Key terms

PPE	Personal protective equipment, such as goggles, gloves and overalls.
Orthographic projection	Drawing an object as separate front, side and plan views.
Isometric drawing	A 3D view in which the receding lines are drawn at 30° to the horizontal.
Tolerance	The acceptable range of size for a finished part.
Specification	A list of requirements that a product must meet.
Prototype	A first working model used to test an idea.
Costing	Working out the cost of materials and labour to set a fair price.
Quality control	Checking that work meets the required standard.

In the workshop

- Producing neat technical drawings
- Measuring twice, cutting once
- Using hand and power tools safely
- Joining by welding, fixing and gluing
- Following recipes and food hygiene rules
- Finishing and presenting work

Study smarter

1. Keep a skills logbook with photos of every piece of work.
2. Practise drawing the three views of everyday objects.
3. Clean and return tools at the end of every lesson.
4. Ask for feedback and improve the next piece.

Check yourself

1. Which three views does an orthographic drawing show?

2. At what angle are the receding lines in an isometric drawing?

3. Name two items of PPE needed for welding.

Answers 1. The front view, the side (end) view and the plan (top) view. · 2. 30° to the horizontal. · 3. Any two of: a welding helmet or mask, leather gloves, a protective apron and closed safety shoes.

Where it leads: Engineering technician, Fabrication and welding, Draughting and architecture, Catering and baking business, Interior design, Beauty therapy business